

ST PETERSBURG PULKOVO, Russian Federation

WMO: 260630

Lat: 59.800N

Lon: 30.263E

Elev: 79

StdP: 14.65

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-9.2	-3.6	-14.6	2.5	-8.7	-8.5	3.5	-2.9	23.2	30.5	20.9	29.9	4.7	100	0.594

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	83.1	67.3	79.4	65.7	76.4	64.1	69.6	79.4	67.5	76.4	65.6	73.9	7.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.1	96.4	74.6	64.2	90.1	72.2	62.4	84.5	70.2	33.6	79.7	31.9	76.2	30.5	73.9	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.2	17.5	15.4	DB	-14.6	88.1	7.2	4.1	-19.8	91.1	-24.0	93.5	-28.1	95.8	-33.3	98.8	(4)
(5)			WB	-15.0	73.2	7.2	2.4	-20.1	75.0	-24.3	76.4	-28.4	77.7	-33.6	79.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.2	21.5	21.0	28.8	41.1	51.6	59.5	65.2	62.2	53.3	42.2	32.7	26.1
	DBStd	17.55	11.79	12.25	8.50	7.66	7.91	6.31	5.77	5.46	6.22	6.93	8.47	10.74
	HDD50	4269	884	812	658	283	75	3	0	0	38	254	520	742
	HDD65	8454	1349	1232	1123	716	421	187	68	121	353	707	970	1207
	CDD50	1427	0	0	0	17	125	289	470	378	136	12	0	0
	CDD65	135	0	0	0	0	5	23	72	34	1	0	0	0
	CDH74	1226	0	0	0	3	71	236	623	284	9	0	0	0
	CDH80	281	0	0	0	0	12	52	160	57	0	0	0	0
Wind	WSAvg	7.0	7.6	7.5	7.5	7.0	6.8	6.5	5.9	6.0	6.3	7.1	7.7	8.2
Precipitation	PrecAvg	25.6	1.8	1.4	1.3	1.3	2.0	2.8	3.1	3.2	2.2	2.4	2.3	1.9
	PrecMax	33.3	3.1	3.5	2.4	2.9	4.8	5.6	6.3	7.2	6.1	4.2	4.4	3.6
	PrecMin	18.5	0.7	0.1	0.3	0.2	0.4	1.0	0.5	0.9	0.6	0.8	0.0	0.6
	PrecStd	3.9	0.6	0.7	0.6	0.7	1.0	1.1	1.4	1.6	1.5	0.9	1.0	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.5	40.8	53.4	71.8	80.9	85.6	89.3	85.7	75.0	61.2	50.1	46.7
		MCWB	39.0	37.9	43.1	55.9	63.7	68.2	70.5	67.7	63.4	54.8	47.8	44.8
	2%	DB	37.8	38.8	45.8	64.5	75.1	80.7	84.4	80.7	69.8	57.2	47.6	42.9
		MCWB	36.5	36.6	39.2	51.2	60.2	66.1	68.6	65.9	61.0	53.2	45.8	41.3
	5%	DB	36.0	37.1	41.4	58.9	70.2	75.6	80.5	76.9	66.0	54.2	45.8	40.6
		MCWB	34.9	35.3	36.7	48.2	56.7	62.9	67.2	64.5	58.3	51.1	44.2	39.3
	10%	DB	34.9	35.4	39.1	54.0	66.2	71.7	77.0	73.4	62.9	52.1	43.2	38.0
		MCWB	33.9	34.0	35.7	45.5	54.5	60.4	65.4	62.9	57.2	49.3	41.8	36.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.0	38.9	43.8	56.9	66.9	70.2	72.9	71.2	64.9	56.0	48.1	44.8
		MCDB	39.7	39.9	51.0	70.0	78.1	81.4	84.1	81.6	72.1	59.6	49.4	46.4
	2%	WB	36.7	36.6	39.8	52.7	62.0	67.4	70.5	67.5	61.9	53.8	45.7	41.2
		MCDB	37.5	37.8	43.8	62.1	70.6	77.6	80.7	75.9	67.6	56.6	46.9	42.4
	5%	WB	35.1	35.1	37.6	49.5	58.7	64.7	68.7	65.7	59.8	52.0	43.9	38.9
		MCDB	35.8	36.3	40.5	57.5	67.5	73.5	77.8	74.3	64.5	54.5	45.1	39.9
	10%	WB	33.7	33.8	35.9	46.0	55.6	62.0	66.7	64.0	57.9	50.0	41.9	36.7
		MCDB	34.4	34.9	38.1	52.4	64.6	69.4	75.1	71.7	62.1	51.9	43.1	37.7
Mean Daily Temperature Range	5% DB	MDBR	9.8	11.4	13.7	16.7	18.9	17.6	17.5	16.9	14.5	10.3	8.1	8.7
		MCDBR	6.3	7.9	16.1	25.0	25.8	24.1	23.1	22.8	19.3	12.9	8.4	7.3
		MCWBR	6.2	7.2	11.2	14.5	13.8	12.6	11.0	10.9	11.5	9.5	8.0	6.6
	5% WB	MCDBR	6.2	8.0	13.1	22.9	22.7	21.2	20.8	19.8	16.5	11.9	8.0	7.1
MCWBR		6.2	7.6	9.6	14.0	13.4	12.1	10.9	10.7	10.8	9.3	7.8	6.7	
Clear-Sky Solar Irradiance	taub	0.277	0.315	0.340	0.380	0.374	0.370	0.396	0.390	0.358	0.336	0.304	0.262	
	taud	2.116	2.130	2.192	2.277	2.379	2.420	2.367	2.376	2.450	2.384	2.209	2.082	
	Ebn at Noon	154	211	247	259	271	274	262	254	244	208	144	112	
	Edh at Noon	14	25	32	35	34	33	34	32	25	20	13	10	
All-Sky Solar Radiation	RadAvg	86	279	741	1189	1613	1691	1634	1281	777	315	88	46	
	RadStd	16	62	98	144	162	124	183	138	75	47	12	7	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.22	N/A	N/A	N/A	N/A	N/A	-436	-509	N/A	N/A
(47) Regional (15 neighbors)		+0.94	N/A	N/A	N/A	N/A	N/A	-328	-324	N/A	N/A

Nomenclature: See separate page